

BCA Semester –I

Subject Title (Theory): Problem solving techniques using ‘C’

Course code	Type of Course	Theory / Practical	Credits	Instructionh our per week	Duration of Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
C1BCA1T1	Paper-1	Theory	04	04	3hrs	20	80	100

Course Outcome (CO): After completion of course (Theory), students will be able to:

CO 1: Familiarize with fundamental concepts and computer programming.

CO 2: Learn fundamental concepts of programming by developing and executing programs in C.

CO 3: Focuses on the structured program.

CO 4: Various constructs and their syntax.

Total Hrs.: 60

Unit I

15 hrs.

Computer Fundamentals: History & Evolution of Computers. Characteristics, Types and Generations of Computers. System logical Organization: Von - Neumann concept of computer with block diagram: Components of Computer & their functions. Input Devices, Output Devices, Storage Devices. Processor & Main Memory: Central Processing Unit: ALU & CU. Architecture of Processor & Main Memory, Processor Registers, Main Memory: Organization of Main Memory, Main Memory Capacity. RAM, ROM, PROM, EPROM, EEPROM, Cache Memory.

Computer Software: Types of Software: System Software & Application Software. Translators: Compiler, Interpreter Linker, Loader and Editor. Computer Languages: Machine Level, Assembly Level & High Level, Their Merits & Demerits. Planning a Computer Program: Algorithm, Flowchart and Pseudo code.

Unit II

15 hrs.

Introduction to C: Over View of C: Introduction. Importance and Features of C. Structure of a C Program. Sample C Programs. Creating and Executing a C Program. Block diagram of execution of C program. Basic Concepts: C Character Set. C tokens: keywords, identifiers, constants and variables. Data types.Declaration & initialization of variables.Symbolic constants. Formatted I/O functions: *printf* and *scanf*: control stings and escape sequences, output specifications with *printf* functions. Unformatted i/o functions to read and display single character and a string: *getchar*, *putchar*, *gets* and *puts* functions.

Operators & Expressions: Arithmetic operators, Relational operators, Logical operators, Assignment operators, increment & decrement operators, bitwise operators, conditional operator and special operators. Computational Problems, Operator Precedence and Associatively.Evaluation of arithmetic expressions, Type conversion.

Unit III

15 hrs.

Control Structures (Branching & Looping): Decision making with *if* statements: *simple if*, *if_else* statements, *nested if_else* and *else_if ladder*. *Switch case* Statement. *goto*, *break&continue* statements. Looping Statements: Entry controlled and Exit controlled, *while*, *do-while&for* loops. Nested loops.

Arrays and Strings: One Dimensional arrays: Declaration, Initialization and Memory representation. Two Dimensional arrays: Declaration, Initialization and Memory representation. Declaring & Initializing string variables. String handling functions: *strlen*, *strcmp*, *strcpy* and *strcat*. Character handling functions: *toascii*, *toupper*, *tolower*, *isalpha*, *isnumeric* etc.

Unit IV

15 hrs.

User Defined Functions: Need for user defined functions. Format of C user defined functions. Components of user defined functions: return type, name, parameter list, function body, return statement and function call. Categories of User defined functions: with and without parameters and return type.

Structures & Unions: Definition of Structure & Union. Declaring structure variables, Accessing structure members, Structure members initialization, Difference between structure and union.

Text Books:

1. P. K. Sinha & Priti Sinha: Computer Fundamentals (BPB)
2. V. Rajaraman: Computer Fundamentals
3. E. Balguruswamy: Programming in ANSI C (TMH)
4. V. Rajaraman: Programming in C (PHI – EEE)
5. Yashwant Kanitkar: Let us C
6. P.B. Kottur: Programming in C (Sapna Book House)

Reference Books:

1. Morismano: Computer Organization & Architecture
2. Norton: Computer Applications
3. Kamthane: Programming with ANSI and TURBO C (Pearson Education)
4. S. Byron Gottfried: Programming with C (TMH)
5. Kernighan & Ritchie: The C Programming Language. (PHI)

Formative Assessment for Theory	
Assessment Occasion/ type	Marks
Internal Assessment Test 1	05
Internal Assessment Test 2	05
Assignment	10
Total	20
<i>Formative Assessment as per guidelines</i>	

Subject Title(Practical):C – Programming Lab

Course code	Type of Course	Theory / Practical	Credits	Instruction hour per week	Duration of Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
C1BCA1P1	Paper-1	Practical	02	04	3hrs	10	40	50

Course Outcome (CO): After completion of course (Practical), students will be able to:

CO 1: Understand the basics of programming by executing the simple programming

CO 2: Be able to design & execution of code.

CO 3: Have practical knowledge of arrays, strings & functions

List of Experiments:

1. Find the area of a circle and area of a triangle given three sides.
2. Largest of three numbers.
3. Reversing the digits of an integer.
4. GCD of two integers.
5. Generating prime numbers.
6. Computing nth Fibonacci numbers.
7. Finding Even and Odd numbers.
8. Exchanging the values of two variables.
9. Counting: Print number from 100 to 200 which are divisible by 7 and display their sum and count using for loop.
10. Summation of set of Numbers.
11. Factorial Computation.
12. Generation of Fibonacci sequence.
13. Array Order Reversal.
14. Finding the Maximum Number in a Set.
15. Removal of Duplicates from an Ordered Array.
16. Partitioning an Array.
17. Finding the k^{th} Smallest Element.
18. Read N (minimum 5) students marks and find number of students passed and fail depending on the marks.
19. Count the number of vowels, consonants and special characters in a given sentence.
20. To find the addition and subtraction of two matrices using function.